

## OLD TOWN 2 W, ME, USA

WMO: 727140

Lat: 44.9281N

Lon: 68.7006W

Elev: 39

StdP: 100.86

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 94644

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -23.3      | -20.2 | -26.9                         | 0.3 | -21.5 | -24.2 | 0.4 | -18.2 | 7.1                   | -9.0 | 6.5 | -8.6 | 1.0                   | N/A  | 0.487 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.8                   | 29.6            | 21.6 | 27.9 | 20.6 | 26.4 | 19.4 | 22.8                | 28.0 | 21.9 | 26.3 | 21.0 | 24.8 | 2.3                  | N/A  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 21.1                            | 15.9 | 24.8 | 20.4 | 15.1 | 23.8 | 19.6 | 14.4 | 23.2 | 67.1          | 28.2 | 64.1 | 26.4 | 60.8 | 24.8 | 25.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |  |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|--|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |  |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |  |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |  |
| 6.1               | 5.3  | 4.5 | DB                         | -27.6 | 32.6               | 4.0 | 0.9                                                | -30.5 | 33.3       | -32.8 | 33.8       | -35.0 | 34.4       | -37.9 | 35.0 |  |
|                   |      |     | WB                         | -27.5 | 24.5               | 2.6 | 0.9                                                | -29.3 | 25.1       | -30.8 | 25.7       | -32.3 | 26.2       | -34.2 | 26.8 |  |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 6.5    | -8.1  | -6.7  | -1.5  | 5.5   | 11.6  | 16.2  | 19.9  | 18.9  | 14.8  | 8.4   | 2.2   | -4.2 |
|                                                                               | DBStd       | 10.67  | 6.79  | 6.05  | 5.45  | 4.08  | 3.65  | 3.34  | 2.71  | 2.80  | 4.01  | 4.18  | 4.68  | 5.72 |
|                                                                               | HDD10.0     | 2317   | 561   | 467   | 357   | 146   | 22    | 1     | 0     | 0     | 5     | 81    | 238   | 439  |
|                                                                               | HDD18.3     | 4461   | 819   | 700   | 614   | 387   | 211   | 81    | 14    | 26    | 119   | 307   | 484   | 698  |
|                                                                               | CDD10.0     | 1038   | 0     | 0     | 1     | 9     | 72    | 187   | 308   | 275   | 149   | 32    | 3     | 0    |
|                                                                               | CDD18.3     | 142    | 0     | 0     | 0     | 0     | 3     | 17    | 64    | 43    | 13    | 1     | 0     | 0    |
|                                                                               | CDH23.3     | 1329   | 0     | 0     | 3     | 6     | 67    | 194   | 564   | 372   | 119   | 6     | 0     | 0    |
|                                                                               | CDH26.7     | 289    | 0     | 0     | 0     | 1     | 16    | 45    | 130   | 72    | 25    | 1     | 0     | 0    |
| Wind                                                                          | WSAvg       | 1.6    | 1.9   | 2.1   | 2.2   | 2.0   | 1.6   | 1.3   | 1.1   | 1.2   | 1.2   | 1.5   | 1.6   | 1.7  |
| Precipitation                                                                 | PrecAvg     | 998    | 78    | 62    | 73    | 74    | 81    | 83    | 82    | 91    | 85    | 95    | 97    |      |
|                                                                               | PrecMax     | 1450   | 217   | 147   | 158   | 172   | 181   | 213   | 196   | 198   | 238   | 181   | 284   | 223  |
|                                                                               | PrecMin     | 664    | 10    | 12    | 5     | 14    | 2     | 27    | 16    | 12    | 17    | 26    | 21    | 34   |
|                                                                               | PrecStd     | 189    | 43    | 31    | 37    | 38    | 37    | 41    | 44    | 43    | 46    | 36    | 49    | 54   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 11.1  | 10.5  | 16.8  | 23.0  | 28.9  | 30.6  | 31.6  | 30.7  | 30.0  | 23.8  | 16.9  | 12.0 |
|                                                                               |             | MCWB   | 10.5  | 8.2   | 10.9  | 15.5  | 18.7  | 21.6  | 22.8  | 22.4  | 22.3  | 16.7  | 15.0  | 11.3 |
|                                                                               | 2%          | DB     | 7.6   | 6.8   | 10.8  | 18.8  | 25.1  | 27.8  | 29.7  | 28.7  | 26.5  | 19.9  | 14.8  | 9.0  |
|                                                                               |             | MCWB   | 6.5   | 3.7   | 6.7   | 12.1  | 16.4  | 20.0  | 21.7  | 21.1  | 19.8  | 16.1  | 12.4  | 7.9  |
|                                                                               | 5%          | DB     | 4.2   | 4.2   | 8.2   | 15.7  | 21.9  | 25.5  | 28.1  | 26.9  | 24.2  | 17.9  | 12.6  | 6.4  |
|                                                                               |             | MCWB   | 2.8   | 2.3   | 4.9   | 9.8   | 14.7  | 18.1  | 20.9  | 20.0  | 19.1  | 15.0  | 9.8   | 4.9  |
|                                                                               | 10%         | DB     | 1.5   | 2.1   | 6.0   | 13.1  | 19.6  | 23.5  | 26.6  | 25.4  | 22.1  | 16.0  | 10.3  | 3.9  |
|                                                                               |             | MCWB   | 0.0   | 0.3   | 3.4   | 8.9   | 13.5  | 16.5  | 19.8  | 18.9  | 17.9  | 13.5  | 8.4   | 2.6  |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 10.3  | 10.0  | 14.2  | 15.7  | 20.7  | 22.4  | 24.6  | 23.6  | 22.9  | 19.0  | 15.2  | 11.2 |
|                                                                               |             | MCDB   | 10.8  | 10.6  | 21.6  | 19.4  | 26.5  | 28.5  | 30.1  | 28.6  | 29.1  | 20.4  | 16.0  | 11.6 |
|                                                                               | 2%          | WB     | 6.9   | 5.2   | 8.3   | 13.3  | 18.0  | 20.9  | 23.0  | 22.4  | 21.7  | 17.2  | 12.9  | 8.5  |
|                                                                               |             | MCDB   | 7.5   | 6.8   | 10.6  | 15.3  | 22.7  | 25.3  | 28.3  | 26.8  | 25.6  | 18.6  | 13.9  | 9.0  |
|                                                                               | 5%          | WB     | 3.5   | 2.8   | 5.3   | 11.1  | 16.4  | 19.3  | 22.1  | 21.3  | 20.4  | 16.0  | 10.1  | 5.6  |
|                                                                               |             | MCDB   | 4.6   | 4.4   | 7.6   | 13.7  | 19.8  | 23.2  | 26.8  | 25.1  | 23.5  | 17.7  | 11.8  | 6.5  |
|                                                                               | 10%         | WB     | 0.8   | 0.8   | 3.1   | 9.1   | 14.7  | 18.0  | 21.2  | 20.5  | 19.2  | 14.2  | 8.1   | 3.0  |
|                                                                               |             | MCDB   | 1.8   | 2.1   | 5.1   | 12.2  | 18.4  | 21.6  | 25.3  | 23.9  | 21.7  | 16.1  | 9.9   | 4.1  |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 10.6  | 11.3  | 10.6  | 11.4  | 12.0  | 11.6  | 11.8  | 12.0  | 12.3  | 10.6  | 9.1   | 9.1  |
|                                                                               |             | MCDBR  | 11.5  | 11.5  | 13.0  | 17.2  | 16.8  | 15.1  | 13.6  | 13.7  | 13.9  | 13.2  | 11.9  | 11.4 |
|                                                                               |             | MCWBR  | 11.4  | 10.2  | 9.3   | 9.6   | 8.8   | 8.1   | 6.8   | 6.8   | 7.6   | 8.8   | 9.9   | 10.1 |
|                                                                               | 5% WB       | MCDBR  | 11.8  | 12.0  | 12.1  | 11.9  | 13.1  | 12.9  | 11.7  | 11.3  | 11.6  | 10.3  | 10.9  | 10.6 |
|                                                                               |             | MCWBR  | 11.7  | 10.6  | 9.4   | 8.4   | 8.2   | 7.7   | 6.4   | 6.1   | 7.3   | 8.2   | 9.8   | 10.4 |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.277  | 0.288 | 0.315 | 0.345 | 0.371 | 0.408 | 0.423 | 0.399 | 0.361 | 0.327 | 0.305 | 0.284 |      |
|                                                                               | taud        | 2.454  | 2.402 | 2.450 | 2.427 | 2.394 | 2.290 | 2.270 | 2.334 | 2.426 | 2.529 | 2.552 | 2.505 |      |
|                                                                               | Ebn at Noon | 851    | 904   | 923   | 917   | 898   | 861   | 843   | 852   | 860   | 850   | 813   | 807   |      |
|                                                                               | Edh at Noon | 69     | 88    | 97    | 108   | 116   | 129   | 130   | 118   | 99    | 78    | 64    | 61    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 1.56   | 2.44  | 3.64  | 4.52  | 5.08  | 5.50  | 5.67  | 5.08  | 4.12  | 2.57  | 1.58  | 1.20  |      |
|                                                                               | RadStd      | 0.16   | 0.23  | 0.24  | 0.47  | 0.62  | 0.43  | 0.45  | 0.30  | 0.26  | 0.24  | 0.15  | 0.11  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (30 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +54     | N/A     |

Nomenclature: See separate page